

06 - Outcomes and Time Gained

Companion reading handout for simultaneous listening.

Course: Medical Prevention of Spontaneous Early Delivery

The phrase "preterm birth" hides several different outcomes.

A trial can ask whether birth occurred before thirty-seven weeks. It can ask whether birth occurred before thirty-four weeks. It can ask about thirty-three, thirty-two, thirty, or twenty-eight weeks. It can restrict the outcome to spontaneous preterm birth. It can include medically indicated deliveries. It can measure neonatal morbidity. It can measure time from diagnosis to delivery. It can measure mean gestational age at birth.

These are not interchangeable.

Threshold Outcomes

Before thirty-seven weeks is the broad boundary. It tells us whether the pregnancy reached term. But it groups together late preterm birth and extreme preterm birth. From the perspective of neonatal risk, that is a very wide container.

Early and Extreme Preterm Thresholds

Before thirty-four weeks is closer to the severe-preterm question. Many progesterone trials and meta-analyses emphasize it because neonatal morbidity becomes more consequential as gestational age falls.

Before twenty-eight weeks asks an even sharper question: did we prevent birth in the extreme-preterm zone? The absolute number of events may be small, so studies can be underpowered, but clinically the endpoint is powerful.

Now ask a different question. What if a treatment does not prevent preterm birth by the under-thirty-seven definition, but it moves delivery later?

Imagine delivery at twenty-eight weeks instead of twenty-six weeks. Still preterm. Still counted as a failure in a binary under-thirty-seven endpoint. But the fetus has gained two weeks. That can be profoundly meaningful.

This is why time-based outcomes matter.

Time-to-Event and Mean Gestational Age

Fonseca did not only report spontaneous delivery before thirty-four weeks. It also used Kaplan-Meier analysis, with gestational age as the time scale. The hazard ratio for continued pregnancy without spontaneous delivery before thirty-four weeks was zero point five seven. That is a time-to-event way of saying the treated group was less likely to deliver spontaneously before the early-preterm threshold across the observed time period.

The PREGNANT trial also used survival-style thinking and a clinically early endpoint, preterm birth before thirty-three weeks. Its primary result was eight point nine percent with vaginal progesterone versus sixteen point one percent with placebo. That is a threshold endpoint, but it is a threshold chosen closer to the severe-preterm problem than thirty-seven weeks.

The Romero two thousand eighteen individual-patient-data meta-analysis is especially useful because it gives us both kinds of language. It reported fewer births before thirty-three weeks: thirteen point one percent with vaginal progesterone versus twenty-one point nine percent with placebo, relative risk zero point six two. It also reported that mean gestational age at delivery was zero point seven four weeks higher in the progesterone group.

Zero point seven four weeks is about five days.

Do not dismiss that because it is less than a week. A mean difference is an average across a population. Some patients may gain little; some may gain more. The result tells us that gestational age at delivery shifted later. It answers a question that a binary endpoint cannot fully answer: did the pregnancy last longer?

Late-initiation studies make this even more important.

If a cervix is first found short after twenty-four weeks, the question is often not only, "Can we prevent birth before thirty-seven weeks?" The question is, "Can we gain time from diagnosis to delivery?" If the patient is diagnosed at twenty-nine weeks, days matter. If treatment does not move the diagnosis-to-delivery interval, then its clinical value becomes harder to defend.

Dominsky's twenty twenty-six study directly addresses this. In the matched analysis of women diagnosed after twenty-four weeks with no prior preterm birth, the interval from diagnosis to delivery was forty-eight point six days with vaginal progesterone and forty-nine point five days without. That is not a signal of prolongation.

The twenty twenty-six American Journal of Obstetrics and Gynecology Maternal-Fetal Medicine abstract also used time-to-delivery analysis. Kaplan-Meier analysis showed no difference, with a log-rank P value of point sixty. The Cox hazard ratio was one point two eight, with a ninety-five percent confidence interval from zero point six five to two point five four. Again, no clear evidence of time gained.

This is the endpoint discipline the course requires.

How to Read an Endpoint

When a paper says the primary outcome was spontaneous preterm birth before thirty-seven weeks, hear that as one choice among many. Ask whether earlier thresholds were measured. Ask whether neonatal outcomes changed. Ask whether gestational age at delivery shifted. Ask whether diagnosis-to-delivery interval changed. Ask whether the endpoint included indicated deliveries or only spontaneous ones.

Endpoint choice is not a technicality. It is a statement of what the investigators considered success.

This also changes counseling.

If a patient asks, "Will this prevent preterm birth?" the scientifically honest answer may be, "The best evidence in patients like you shows a reduction in birth before a specific early threshold." Or, "The evidence does not show that starting this medication now prolongs pregnancy." Or, "The trial reduced a birth threshold but was not powered to prove neonatal morbidity reduction."

That is more careful than saying yes or no. It is also more useful.

Store the endpoint hierarchy.

Under thirty-seven weeks tells us term versus preterm, but it is broad. Under thirty-four and under-thirty-three weeks are more directly connected to early preterm prevention trials. Under twenty-eight weeks captures extreme prematurity. Gestational age at delivery and diagnosis-to-delivery interval tell us about time gained. Neonatal composite outcomes tell us whether the infant outcome changed, not only whether a threshold was crossed.

A good physician should not be satisfied with the phrase "reduced preterm birth." A good physician asks: which preterm birth, by what definition, and did the fetus gain meaningful time?